

SUPERBatch

Flexible Batch Management Software based on ISA S88 Standard



SUPERBatch effectively manages flexible multi-stream and multi-product batch operations found in process industries including specialty chemicals, life sciences, food & beverage, consumer packaged goods, paints, lube oils etc. SUPERBatch integrates seamlessly with **EcoStruxure™ Process Expert** and **EcoStruxure™ Automation Expert**, enabling a future-ready, scalable automation architecture on **Schneider Electric** platforms.

Benefits

- Consistent quality to recipe specification
- Increase Asset Utilization
- Increase Operational Efficiency
- Less dependence on Human Skills
- Create and modify recipes with your own process experts
- Reduction in Batch Cycle Time
- Reduction in Raw Material Wastage
- Reduction in Energy Consumption
- Automatic Batch Reports
- Comparison with Golden Batch
- Reduce cost of regulatory compliance
- Ready for Integration with SAP

SUPERTECH

INSTRUMENTATION SERVICES (I) PVT. LTD.

Master Alliance Partner

Life Is On

Schneider
Electric

ACHIEVE CONSISTENT QUALITY

SUPERBatch helps you achieve consistent product quality by automating concurrent execution of batch processes in multiple equipment

MAXIMIZE ASSET UTILIZATION

Batch scheduling, equipment arbitration, allocation and release helps maximum utilization of assets.

MATERIAL MANAGEMENT & TRACEABILITY

Plant Editor with material management enables the batch engine to manage flexible product paths, simultaneous batch execution while documenting material flows and traceability records

FLEXIBLE RECIPE MANAGEMENT

Process supervisors without any knowledge of control systems will be able to create new recipes and modify existing ones in a user friendly SOP type format.

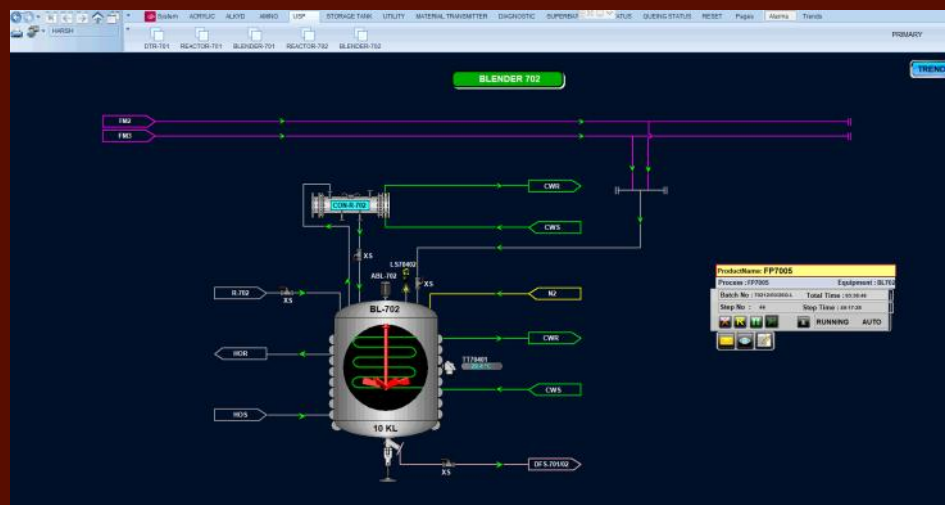
ELECTRONIC BATCH RECORD (EBR)

SUPERBatch offers comprehensive batch execution and equipment history with full product genealogy and material traceability through automatic electronic records to the historical database. It also helps in regulatory compliance.

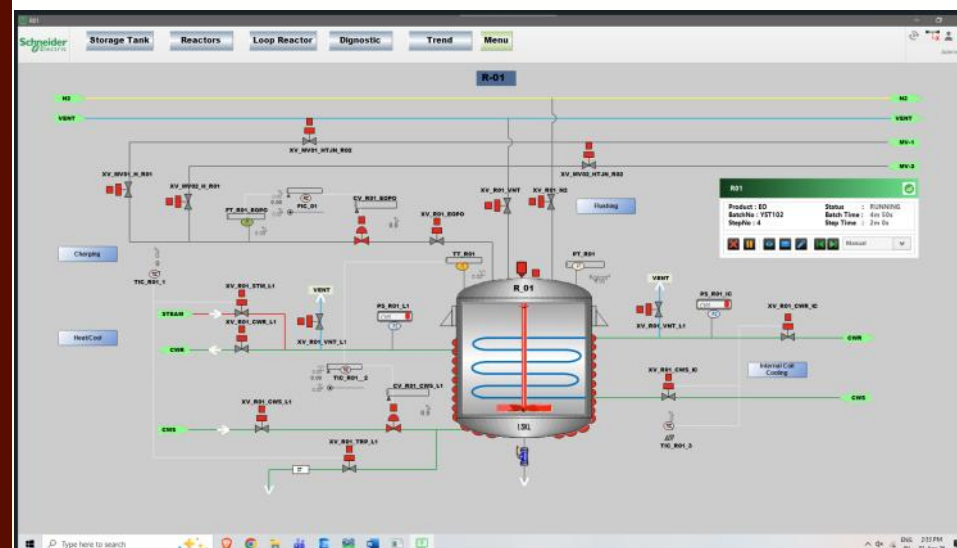
INTEGRATION WITH ERP

Batch system is ready for integration with an ERP system such as SAP

SUPERBatch for a Resin Plant with EcoStruxure Process Expert



SUPERBatch for an Ethoxylation Plant with EcoStruxure Automation Expert



On-line Batch Viewer

Online Batch Viewer

BATCH HEADER

BATCH NO

10000

PRODUCT CODE

ED

EQUIPMENT

R01

BATCH SIZE

2000

DATE & TIME

01 APR 2016 10:07:23 AM

CONNECTION

ONLINE

BATCH DETAILS

STEP NO	START DATE TIME	END DATE TIME	MESSAGE	TRANSITION CRITERIA	STD TIME	MATERIAL	STD QTY	ACT QTY	STD TEMPERATURE	ACT TEMPERATURE	STD PRESSURE	ACT PRESSURE	STD FLOW	ACT FLOW
1	01 Apr 2016 10:07:23 AM	01 Apr 2016 10:07:23 AM	Batch number entry	Operator Action	5m		0	0	25	24	0	0	0	0
2	01 Apr 2016 10:07:23 AM	01 Apr 2016 10:07:23 AM	Check Reactor is clean, empty & bottom valve closed	Manual Confirmation	5m		0	0	25	24	0	0	0	0
3	01 Apr 2016 10:07:23 AM	01 Apr 2016 10:07:23 AM	Utility availability verification	Manual Confirmation	5m		0	0	23	23	0	0	0	0
4	01 Apr 2016 10:07:23 AM	No Date	Manual raw material charging	Manual Material Charge	30m	W001	100	100	25	25	0	0	0	0
5	No Date	No Date	Nitrogen Purging (3 cycles)	Flushing	50m		0	0	25	0	0.2	0	-0.1	0
6	No Date	No Date	Reactor heating to 100 °C	Heating	1h		0	0	100	0	0	0	0	0
7	No Date	No Date	Start ED addition with flow control	Auto Material Charge	2h	ED	1800	0	100	0	0	0	0	0
8	No Date	No Date	Condensate draining from top coil	Drinking	5m		0	0	100	0	0	0	0	0
9	No Date	No Date	Start ED addition (PIC-05)	Auto Material Charge	1h	ED	1000	0	100	0	0	0	0	0
10	No Date	No Date	Start cooling through internal coil	Cooling	30m		0	0	80	0	0	0	0	0

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SUPERBatch Modules

Batch Server

SUPERBatch Manager



- Defines various security levels
- Provides user based authenticity for each module of SUPERBatch

Plant Editor



- Defines Plant Tree showing all plant components
- Create Masters— Materials, Quality, Process Parameters, Operator Guide Messages

Control Modules

Recipe Engine



- Defines various security levels
- Provides user based authenticity for each module of SUPERBatch

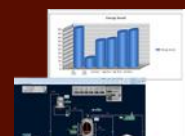
Batch Scheduler



- Batch Registry
- Generation of Control Recipes.
- Scheduling of Batches.

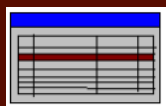
Execution Modules

Batch Execution



- Execute authentic scheduled batches
- Provision for Downloading control recipe for Batch execution

On-line Batch View



- Shows current status of the batch with data for completed steps

On-line Recipe Change



- Online modifications in Control Recipe during Batch execution.

Batch Portal



- On-line Batch View
- Standard Reports
- Customized Reports

SUPERBatch Manager is used to create user profiles and apply security levels to avoid security risks

Plant Editor Module of SUPERBatch is used for defining and storing the information of all plant elements namely Main Processing Units, Equipment's, Processing Actions, Connecting Elements, Process Parameters, and Operator Guide Messages.

Recipe Engine in SUPERBatch is used for Recipe Management which performs tasks like Creation of New Master Recipes, Modification, Deletion of master recipes etc. Recipe Engine consists of two parts viz. Recipe Header and SOP.

Recipe Header consists of Recipe information like Id (Identification Number), Name, Recipe Version, Date of Creation, Creator Name, Product for which Recipe is getting designed, Main Processing Unit for which Recipe is getting defined, Minimum, Maximum Batch Size. SOP is the Sequence of Process is defined by adding each and every individual step for the process. The values of the process parameters, Operator guide Message, Time required for execution of step, Equipment, Processing Action and status of the corresponding relevant connecting elements like valves, pumps are to be defined during each step definition.

Batch Scheduler allows you to create Control recipes and schedule the batches based on the resources available.

Batch Execution module allows you to execute the scheduled batches that will run according to the control recipes.

On-line Batch View allows you to view the step-wise status of each batch giving details of the completed steps.

On-line Recipe Changer allows you to modify the recipes for the unfinished steps during Batch execution

Batch Portal allows on-line viewing of Batch Status and reports on Web Browser from any PC, Laptop or Mobile connected on the plant network.

Various reports such as the Batch Report (Electronic Batch Record), Raw Material Variance Report, Temperature profile over the Batch period— Comparison with Golden Batch, Batch to Batch comparison against key performance indicators such as Batch Cycle time, Yield, Quality parameters, raw material consumption are automatically generated. Various customized reports can also be generated as per client's requirement.

Licensing

1. 6 Nos. Equipment Server Paper License: SIS-SB-SR-S
2. 15 Nos. Equipment Server Paper License: SIS-SB-SR-M
3. Unlimited Equipment Server Paper License: SIS-SB-SR-L
4. Client Paper License: SIS-SB-CL

INDUSTRIES SERVED

- Industrial Resins— Alkyd, Acrylic, Amino, Polyester, CED, Phenolic
- Paints—Decorative, Industrial, Automotive, Marine
- Adhesives
- Inks
- Surfactants—Ethylene Oxide Based
- Additives—Food, Polyester Films
- Agrochemicals
- API (Pharma)
- Food & Beverages—Liquid Food, Chocolates, Beverages
- Flavours & Fragrances
- Industrial Lubricants
- Consumer Packaged Goods
- Dyes, Pigments
- Solvent Extraction
- Animal Feeds



SUPERTECH is a leading “**TURNKEY SOLUTIONS PROVIDER FOR AUTOMATION, INSTRUMENTATION & ELECTRICAL**”. With a successful track record of over 41 years, Supertech undertakes Turnkey Automation, Instrumentation & Electrical Projects all over India and abroad in a wide range of Process Industries such as Industrial Resins, Paints, Polymers, Specialty Chemicals, CPG Lube Oil Blending, Food & Beverage, Oil & Gas and Agrochemicals.

SUPERTECH

INSTRUMENTATION | AUTOMATION |
ELECTRICAL | CONTRACTING

Master Alliance Partner with **Schneider Electric** for

- Control Systems (PLC/ SCADA)
- EcoStruxure Process Expert (DCS)
- EcoStruxure Automation Expert (New Gen DCS)
- CPG Segment Specialist
- Energies & Chemicals Segment Specialist
- Modicon Safety PLC Specialist
- Aveva Registered System Integrator
- EcoXpert - Energy Monitoring Solutions
- Intelligent MCC, VFD, Soft Starters

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